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004606

May 2025

B.Tech. (CSE) (AI & ML) (Sixth Semester)

Distributed System (PEC-CS-S-602 (I))

Time : 3 Hours] [*Maximum Marks : 75*

Note : It is compulsory to answer all the questions (1.5 marks each) of Part A in short. Answer any *four* questions from Part B in detail. Different sub-parts of a question are to be attempted adjacent to each other.

Part A

1. (a) Define horizontal fragmentation, vertical fragmentation and hybrid fragmentation in a DDBMS. **1.5**

   (b) Discuss the concept of semi-join and its role in distributed query processing. **1.5**

   (c) Compare entity integrity, referential integrity and semantic integrity. **1.5**

   (d) What are the trade-offs between availability, consistency and performance in distributed design ? **1.5**

(e) Compare cost-based and rule-based query optimization. Which is more effective in distributed databases ? 1.5

(f) Compare the pessimistic and optimistic transaction models. 1.5

(g) Discuss, how dynamic query optimization differs from static query optimization. 1.5

(h) Compare symmetric fragmentation and asymmetric fragmentation approaches in query optimization. 1.5

(i) Differentiate between transaction failure, system failure, media failure and communication failure. 1.5

(j) Compare synchronous and asynchronous commit protocols in distributed transactions. 1.5

Part B

2. (a) Define and explain the following transparencies in a DDBMS: Location Transparency, Replication Transparency, Fragmentation Transparency. How do these transparencies improve the usability and efficiency of a distributed database system ? 7.5

(b) What role do protocols like Two-Phase Commit (2PC) and Three-Phase Commit (3PC) play in ensuring distributed transaction consistency ? 7.5

3. (a) How does the CAP theorem (Consistency, Availability and Partition Tolerance) influence the choice of distributed database design ? 7.5

(b) What is semantic integrity control and why is it more complex in a distributed database environment compared to a centralized system ? 7.5

4. (a) What is derived fragmentation ? Compare primary horizontal fragmentation and derived horizontal fragmentation. 7.5

(b) Differentiate between strict, rigorous and optimistic concurrency control techniques. 7.5

5. (a) How does the ACID model define transactions ? Explain the ACID properties and their significance in transaction processing. 7.5

(b) What are the differences between immediate update recovery and deferred update recovery ? 7.5

6. (a) What is shadow paging and how does it contribute to database reliability ? 7.5

(b) Explain, how the wait-die and wound-wait schemes prevent deadlock in distributed transactions. 7.5

7. (a) What is a deadlock, and how does it occur in database systems ? Compare deadlock prevention and deadlock avoidance techniques. 7.5

(b) How do two-phase locking (2PL) and timestamp ordering ensure concurrency control in centralized databases ? 7.5

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Roll No. .....

018603

May 2024

B.Tech. (CSE(AIML)) (Sixth Semester)

Distributed Systems (PEC-CS-S-602-I)

Time : 3 Hours] [*Maximum Marks : 75*

Note : It is compulsory to answer all the questions (1.5 marks each) of Part A in short. Answer any *four* questions from Part B in detail. Different sub-parts of a question are to be attempted adjacent to each other.

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Part A

1. (a) Write advantages of Distributed DBMS. **1.5**

   (b) Define query decomposition. **1.5**

   (c) Define Data Security. **1.5**

   (d) What do you mean by Transitive Dependency ? **1.5**

   (e) Name the layers of query processing. **1.5**

   (f) Name one concurrency control algorithm. **1.5**

   (g) What are ACID properties ? **1.5**

(h) Differentiate homogeneous and heterogeneous databases. 1.5

(i) Differentiate between fault and failure. 1.5

(j) Define Reliability. 1.5

Part B

2. What are distributed database systems ? What are transparencies in distributed DBMS ? Explain them in detail. 15

3. (a) What is query processing ? What are its objectives ? 10

   (b) Discuss Fragmentation. 5

4. What are the goals of transaction management ? What are different models of transactions ? Explain any *two* of them in detail. 15

5. (a) What are reliability issues in DDBSs ? What are different types of failures ? 10

   (b) What are disadvantages of DDBS ? 5

(h) Differentiate homogeneous and heterogeneous databases. 1.5

(i) Differentiate between fault and failure. 1.5

(j) Define Reliability. 1.5

Part B

2. What are distributed database systems ? What are transparencies in distributed DBMS ? Explain them in detail. 15

3. (a) What is query processing ? What are its objectives ? 10

   (b) Discuss Fragmentation. 5

4. What are the goals of transaction management ? What are different models of transactions ? Explain any *two* of them in detail. 15

5. (a) What are reliability issues in DDBSs ? What are different types of failures ? 10

   (b) What are disadvantages of DDBS ? 5

6. (a) How Deadlock is managed in Distributed Systems ? 10

(b) Discuss Semantic Integrity Control. 5

7. Explain in detail 2-phase Commit protocol and 3-phase Commit protocol of Distributed Reliability Protocols. 15

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004604

May 2023

B.Tech. (CSE/CSE(AIML)) 6th Semester

Distributed Systems (PEC-CS-S-602-I)

7/25

Time : 3 Hours] [Max. Marks : 75

Instructions:

1. *It is compulsory to answer all the questions (1.5 marks each) of Part-A in short.*
2. *Answer any four questions from Part-B in detail.*
3. *Different sub-parts of a question are to be attempted adjacent to each other.*

PART-A

1. (a) Differentiate between DBMS and Distributed DBMS. (1.5)

   (b) Explain Horizontal Fragmentation. (1.5)

   (c) Write various anomalies caused by concurrency. (1.5)

   (d) Differentiate between Strong Coupling and Weak Coupling. (1.5)

   (e) Explain properties for 2NF and 3NF. (1.5)

   (f) Discuss various failures in Distributed DBMS. (1.5)

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(g) What do you mean by Affinity Matrix. (1.5)

(h) Differentiate between fault and failure. (1.5)

(i) Explain Shared-Nothing architecture of Parallel Database systems. (1.5)

(j) What are the different states of Transaction? (1.5)

PART-B

2. (a) Explain architecture of Distributed Database using Top-Down approach. (10)

(b) Explain Data Partitioning methods in Parallel Database Systems. (5)

3. (a) Explain ACID properties of Transactions in detail with the help of an example. (5)

(b) Explain in detail all layers of Query Processing. (10)

4. Explain in detail OSI reference Model. (15)

5. (a) What do you mean by in-place recovery in Distributed System. Explain with an example. (5)

(b) What do you mean by Concurrency? Explain three Timestamp based Concurrency Control strategies in detail. (10)

6. What are the various kinds of transparencies in distributed database design? Explain each with the help of example. (15)

7. Explain in detail 2-phase Commit protocol of Distributed Reliability Protocols. (15)